

## Questions – Mendelian Randomization Exercise

1. Is there significant evidence of an association between LRRC37A2 and sensorineural hearing loss? \_\_\_\_\_

2. Is there any evidence that some of the variants have horizontal pleiotropy? \_\_\_\_\_

3. How many variants? \_\_\_\_\_

4. What is the p-value, beta, and standard error before outlier removal? These are the results from the IVW test.

A. P-value: \_\_\_\_\_

B. Beta: \_\_\_\_\_

C. Standard Error\*: \_\_\_\_\_

5. What is the p-value, beta, and standard error after outlier removal? Removal of the variants that display horizontal pleiotropy.

A. P-value: \_\_\_\_\_

B. Beta: \_\_\_\_\_

C. Standard Error\*: \_\_\_\_\_

6. Why is the p-value smaller after outlier removal?

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\_\_\_\_\_  
\_\_\_\_\_

7. Does an increase in LRRC37A2 protein expression levels increase or decrease the risk of developing sensorineural hearing loss? \_\_\_\_\_

8. What is the potential underlying mechanism which is suggested by the results:

A) The risk of sensorineural hear loss is due to genetic variants, which in turn causes LRRC37A2 protein to be released in the blood, for example through hair cell death and/or damage? \_\_\_\_\_

B) Genetic variants are modulating LRRC37A2 protein expression levels in the blood which modifies sensorineural hearing loss risk? \_\_\_\_\_

\*Note MR-PRESSO using sd for the nomenclature for standard error